

Vishnu Vardhan Yarraguntla
Data Engineer
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PROFESSIONAL SUMMARY

Data Engineer with 3+ years of experience designing scalable cloud data pipelines and modern data warehouses using Azure and Apache Spark technologies. Skilled in building ETL workflows, optimizing large-scale data processing, and implementing dimensional data models for enterprise analytics. Proven track record of processing 5TB+ daily data workloads, improving pipeline performance by 25%, and reducing infrastructure costs by 40% through cloud migration and automation.

TECHNICAL SKILLS

- **Cloud Platforms:** Microsoft Azure (Synapse, Data Factory, Databricks, Blob Storage), AWS (S3, EC2)
- **Big Data Technologies:** Apache Spark (PySpark, Spark SQL), Hadoop (HDFS, Hive), Kafka
- **Programming Languages:** Python, SQL, Scala, Bash
- **Databases & Warehouses:** SQL Server, PostgreSQL, Snowflake, Azure Synapse
- **ETL & Orchestration:** Azure Data Factory, Apache Airflow, Apache NiFi, Cron
- **Data Modeling & BI:** Dimensional Modeling (Star/Snowflake), Data Vault 2.0, ER Modeling, Power BI, Tableau
- **Tools & DevOps:** Git, Docker, Linux, Azure DevOps, Jira, Confluence, Agile/Scrum

Professional Experience

GE Healthcare | Data Engineer | Chicago, IL | Apr 2024 – Present

- Architected and managed Azure data pipelines processing 5TB+ of daily medical device telemetry and patient data, increasing data availability for analytics by 30%.
- Optimized PySpark jobs within Azure Databricks, reducing average pipeline execution time by 25% and improving cluster resource utilization.
- Built dimensional data models within Azure Synapse Analytics to serve 10+ global sales and service dashboards, decreasing report generation time by 20%.
- Implemented data validation frameworks in Azure Data Factory, elevating data accuracy for critical patient metrics to 99.8%.
- Automated the provisioning of modeled datasets for Power BI, saving 15 hours per week in manual preparation for the analytics team.
- Migrated 3 legacy on-premise SQL Server data marts to Azure Synapse Analytics, reducing monthly infrastructure costs by 40%.
- Designed a data governance model for PHI (Protected Health Information), ensuring compliance with HIPAA standards across all data assets.
- Collaborated with data science teams to productionize machine learning models by creating feature stores and inference pipelines in Databricks.
- Established monitoring and alerting for 50+ data pipelines using Azure DevOps, decreasing incident response time by 60%.

Environment: Azure Data Factory, Azure Databricks, Azure Synapse Analytics, PySpark, SQL Server, Power BI, Python, Git, Azure DevOps, Docker.

Biocon | Data Engineer | India | Aug 2021 – Jul 2023

- Consolidated data from 5+ source systems (SAP, LIMS, MES) into a unified data lake, eliminating data silos and improving cross-departmental access.
- Engineered batch processing solutions with PySpark, transforming and aggregating 2TB+ of monthly clinical trial data, which cut processing time by 35%.
- Created and maintained 20+ Hive tables and queries on a Hadoop cluster, enabling historical analysis of 10+ years of R&D data.
- Automated 10+ recurring data extraction and reporting tasks, saving the bioinformatics team 20 hours of manual effort each week.
- Developed 50+ SQL scripts and stored procedures for data cleansing, ensuring 100% adherence to FDA and GxP compliance standards.
- Documented end-to-end data lineage and pipeline specifications, reducing onboarding time for new team members by 50%.
- Standardized ETL processes across the analytics department by creating reusable Python modules and templates, improving development consistency.
- Accelerated data validation for batch release by building an automated reconciliation tool in Python, reducing the review cycle from 2 days to 4 hours.
- Configured and managed Hadoop cluster security and user permissions, implementing role-based access control for sensitive research data.

Environment: Hadoop, Hive, PySpark, Python, SQL, SAP, SSIS, Linux, Windows Server, Tableau, Cron.

EDUCATION

- Master of Science in Information Technology
Trine University — USA
- Bachelor of Engineering in Electrical and Communications Engineering
KKR and KSR Institute of Technology and Science — India